# Karim Khalil

**Computer Vision / AI Engineer**

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## PROFESSIONAL SUMMARY

AI engineer focused on computer vision, with 2+ years of experience developing, training and deploying YOLO-based object detection models for live video and aerial imagery. Experience serving models through FastAPI, containerizing them with Docker, and building the web front ends that present the results. Projects include PPE compliance, restricted-area intrusion, people counting, driver fatigue, fire and smoke detection, and palm counting from aerial imagery.

## TECHNICAL SKILLS

**Computer Vision:** YOLOv8, YOLO26, object detection, OpenCV, real-time video processing, CNNs, activity recognition
**Model Development:** Training and validating models on custom datasets (LabelImg), transfer learning, inference optimization
**ML Frameworks:** Python, PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, Dlib
**Deployment and APIs:** FastAPI, REST APIs, Docker, multi-threading, Postman, Git, GitHub
**Web and Geospatial:** HTML, JavaScript, Leaflet maps, KML farm boundaries, satellite and aerial imagery, NDVI
**AI Tooling:** Claude Code, LLM applications, OpenClaw, Replit, SQL

## PROFESSIONAL EXPERIENCE

**Platform Operation & Data Analyst | Platfarm | Royal Commission for AlUla (RCU)**
Jun 2026 - Present | AlUla, Saudi Arabia (On-site)

- Built Date Palm AI: trained a custom YOLO26 model that counts date palms in aerial imagery for individual farms.
- Managed platform operations and validated dashboards and reports to ensure data accuracy, quality and consistency.
- Built AI applications and automated workflows using Python, Replit, OpenClaw and LLM technologies, including a WhatsApp reporting assistant.

**AI Engineer | Nasam Technology**
May 2025 - Apr 2026 | Raidah, Saudi Arabia (Remote)

- Developed an oil spill detection model using deep learning on satellite and aerial imagery for environmental monitoring.
- Built NDVI analysis workflows to assess crop health and vegetation stress from remote sensing data.
- Carried out fleet data analysis.

**AI Engineer | Omni Clouds**
Jul 2024 - Feb 2025 | Dubai, UAE (Remote)

- Developed and deployed computer vision models for real-time video monitoring, containerized with Docker for internal production use.
- Implemented YOLO-based object detection pipelines for live video streams, improving detection reliability and reducing false alerts.
- Collaborated with software engineers to integrate AI models into internal systems and applications, supporting continuous real-time inference and system stability.

## PROJECTS

**Date Palm Detection & Counting (YOLO26)**

- Trained a custom YOLO26 model to detect and count date palms in aerial and satellite imagery, with an upload page and a farm map showing counts per farm.
- Measures palm size per image and tiles inference to keep counts consistent; counts only palms inside each farm boundary.
- Tools: Python, YOLO26, PyTorch, OpenCV, FastAPI, Leaflet, KML

**PPE Detection System**

- Implemented YOLOv8 object detection models to monitor PPE compliance in industrial environments.
- Trained and validated models on custom datasets.
- Tools: Python, PyTorch, YOLOv8, OpenCV, LabelImg

**Restricted Area Intrusion Detection System**

- Developed an AI security system using object detection and motion analysis for factory surveillance.
- Generated real-time alerts with visual overlays.
- Tools: Python, OpenCV, YOLOv8, NumPy

**People Detection and Counting in Restricted Zones**

- Detected and counted people inside a restricted zone in real time and saved the captured images to a folder.
- Tools: Python, OpenCV, object detection

**Driver Inattention Monitoring System**

- Developed a real-time driver monitoring system using CNNs and facial landmark tracking to detect fatigue and distraction.
- Optimized inference for continuous real-time processing.
- Tools: Python, PyTorch, OpenCV, Dlib, multi-threading, NumPy

**Human Activity Detection System**

- Built activity recognition models using transfer learning on temporal video data for elderly safety and posture monitoring, with real-time inference suitable for continuous alerting.
- Tools: Python, TensorFlow, Keras, OpenCV, Scikit-learn

**Fire and Smoke Detection System**

- Engineered a CNN-based classifier for early fire and smoke detection across varied environments, optimized for low-latency deployment in surveillance pipelines.
- Tools: Python, TensorFlow, Keras, OpenCV, Pandas, NumPy

## EDUCATION

**B.Sc. in Computer Science** | Canadian International College (CIC) | 2024

## CERTIFICATIONS

AI For Everyone - Coursera; Data Analysis - egFWD; GIS Professional - NARSS; Competitive Programming Level 1 - Coach Academy EG; Communication Skills - Canadian International College

## LANGUAGES

Arabic (native), English (professional)